

Online resources — har topic ka OFFICIAL link + “is project me kahan”

Fresher ke liye: pehle column ka concept padho (link official docs ka hai — sabse bharosemand source), doosra column batata hai isi project me woh kahan LIVE use hua hai (real example > tutorial).

Django basics

Topic	Official link	Is project me kahan
Tutorial (start here)	docs.djangoproject.com/en/5.0/intro/tutorial01/	poora project iska adult version hai
Models	docs.djangoproject.com/en/5.0/topics/db/models/	<code>production/models/*.py</code> , <code>expense/models.py</code>
Field types + options	docs.djangoproject.com/en/5.0/ref/models/fields/	DecimalField money (kabhi Float nahi!), JSONField metadata
ForeignKey + on_delete	docs.djangoproject.com/en/5.0/ref/models/fields/#django.db.models.ForeignKey	PROTECT money-anchors (KNOWLEDGE_MAP §6 policy)
Abstract base models	docs.djangoproject.com/en/5.0/topics/db/models/#abstract-base-classes	<code>core/models.py</code> TimeStampedModel
Custom managers	docs.djangoproject.com/en/5.0/topics/db/managers/	<code>core</code> ActiveManager <code>.active</code>
Migrations	docs.djangoproject.com/en/5.0/topics/migrations/	<code>*/migrations/</code> ; data-migration = backfill known facts only

Queries & data

Topic	Link	Yahan
QuerySet API	docs.djangoproject.com/en/5.0/ref/models/queries/	har service ka read code
Aggregation	docs.djangoproject.com/en/5.0/topics/db/aggregation/	<code>payroll_service</code> SUM/Count rollups
F / Q objects	docs.djangoproject.com/en/5.0/topics/db/queries/#complex-lookups-with-q-objects	era filters <code>(Q(settlement_line__isnull=True) Q(...))</code>
<code>select_related</code> / <code>prefetch_related</code>	docs.djangoproject.com/en/5.0/ref/models/queries/#select-related	settlement preview joins; perf-baseline tests
Constraints (Check/Unique)	docs.djangoproject.com/en/5.0/ref/models/constraints/	PSI XOR, finalized⇒settled_at, WST partial-unique

Transactions & locking (paisa-code ka backbone)

Topic	Link	Yahan
Transactions / atomic	docs.djangoproject.com/en/5.0/topics/db/transactions/	har service <code>@transaction.atomic</code>
<code>select_for_update</code>	docs.djangoproject.com/en/5.0/ref/models/queriesets/#select-for-update	finalize/void/consume — lock-then-recheck
PostgreSQL explicit locking	postgresql.org/docs/current/explicit-locking.html	lock ORDER §11.5 samajhne ko
Advisory locks	postgresql.org/docs/current/functions-admin.html#FUNCTIONS-ADVISORY-LOCKS	<code>pg_advisory_xact_lock(5374)</code>
Isolation levels	postgresql.org/docs/current/transaction-iso.html	background theory

Views / URLs / forms / templates

Topic	Link	Yahan
Class-based views	docs.djangoproject.com/en/5.0/topics/class-based-views/	TemplateView/FormView/View pattern everywhere
URL dispatcher	docs.djangoproject.com/en/5.0/topics/http/urls/	har <code>urls.py</code> (headers me route maps)
Forms	docs.djangoproject.com/en/5.0/topics/forms/	<code>production/forms/</code> , expense forms
Templates	docs.djangoproject.com/en/5.0/topics/templates/	<code>accounts/base.html</code> + partials
Messages framework	docs.djangoproject.com/en/5.0/ref/contrib/messages/	har action ka redirect+message
Middleware	docs.djangoproject.com/en/5.0/topics/http/middleware/	SidebarAccessMiddleware

Auth & security

Topic	Link	Yahan
Custom User model	docs.djangoproject.com/en/5.0/topics/auth/customizing/	<code>accounts/models.py</code> email login
Permissions/auth	docs.djangoproject.com/en/5.0/topics/auth/default/	<code>permission_service</code> wraps it
Password hashing (Argon2)	docs.djangoproject.com/en/5.0/topics/auth/passwords/	settings hashers
django-allauth	docs.allauth.org	Google OAuth (pre-provisioned only)
OWASP cheat sheets	cheatsheetseries.owasp.org	rate-limit/auth design background

SQL & PostgreSQL (saath me LEARNING/09 padho)

postgresql.org/docs/current/tutorial.html · use-the-index-luke.com (indexes ki BEST free book) · postgresql.org/docs/current/sql-explain.html (EXPLAIN).

Python (jahan atakō)

docs.python.org/3/tutorial/ · `decimal` module (paise INT/Decimal me, float kabhi nahi):
docs.python.org/3/library/decimal.html · `pathlib`, `re` refs.

Frontend (server-rendered yahan)

MDN HTML/CSS/JS: developer.mozilla.org · CSS grid/flex: css-tricks.com/snippets/css/complete-guide-grid/ ·
DataTables: datatables.net/manual/ (list pages) · Paged print/media queries: MDN @media.

Architecture patterns (yeh project kyu aisa hai)

Concept	Link	Yahan
Service layer (a.k.a. application services)	martinfowler.com/eaCatalog/serviceLayer.html	ADR-0001; <code>*/services/</code>
Ledger / append-only accounting	martinfowler.com/eaDev/AccountingNarrative.html	WorkerLedgerEntry
Event sourcing-ish corrections	martinfowler.com/eaDev/EventSourcing.html (background)	reverse/supersede chains
ADRs (decision records)	adr.github.io	<code>docs/adr/</code>
Open-closed principle	en.wikipedia.org/wiki/Open%E2%80%93closed_principle	StageHandler engine
Strangler-fig migrations	martinfowler.com/bliki/StranglerFigApplication.html	V2 dual-write → cutover

Seekhne ka tareeka (fresher steps)

1. Tutorial part 1–4 (Django) → 2. Is project ka KNOWLEDGE_MAP + LEARNING 01/03/04 → 3. Ek chhota read-only view khud trace karo (My Earnings) → 4. LEARNING/09 SQL levels 1–3 dbshell pe → 5. Phir paanch core service files. Link tab kholo jab project ka code samajh na aaye — context ke saath padhna 10× tez hai.